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Chapter: Number of Provinces (LC 547)

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Algorithm Notes & Logic

- **DSU Setup:** *initially sab apne apne baap he*
- **Methodology:** *iss baar dsu se kr rha*
- **Initialization:** *initially har city ek alag group*
- **Processing:** *phir jo bhi connected he unka union*
- **Final Result:** *baad me jitne unique parent utne provinces*

Disjoint Set Union (DSU) Implementation

```
class dsu {
public:
    vector<int> parent;
    vector<int> size;

    dsu(int n) {
        parent.resize(n);
        size.resize(n);

        for(int i = 0; i < n; i++) {
            parent[i] = i; // initially sab apne apne baap he
        }
    }

    int findparent(int node) {
        if(parent[node] == node) {
            return node;
        }

        parent[node] = findparent(parent[node]);
        return parent[node];
    }
};
```

```

    }

    void unionit(int u, int v) {
        int pu = findparent(u);
        int pv = findparent(v);

        if(pu == pv) return;

        if(size[pu] < size[pv]) {
            swap(pu, pv);
        }

        parent[pv] = pu;
        size[pu] += size[pv];
    }
};

```

Solution Approach

```

class Solution {
public:
    int findCircleNum(vector<vector<int>>& isConnected) {

        // iss baar dsu se kr rha
        // initially har city ek alag group
        // phir jo bhi connected he unka union
        // baad me jitne unique parent utne provinces

        int n = isConnected.size();
        dsu ds(n);

        for(int i = 0; i < n; i++) {
            for(int j = 0; j < n; j++) {
                if(isConnected[i][j] == 1) {
                    ds.unionit(i, j);
                }
            }
        }

        unordered_set<int> st;
        for(int i = 0; i < n; i++) {
            st.insert(ds.findparent(i));
        }
    }
};

```

```
        return st.size();  
    }  
};
```